Ex.NO:- 6	Social Media Advertisement Click Rate Analysis
Date:-	

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Aim:-

To analyze advertisement performance data using statistical hypothesis testing and correlation analysis to determine whether two advertisement formats have significantly different click-through rates and to study relationships between user age, viewing time, and click behavior.

Problem Definition

A company tested two advertisement formats (Format A and Format B) and collected data on:

- Total impressions
- Total clicks
- User age
- Viewing time
- Click behavior

Objectives:

1. Test whether there is a significant difference in CTR between the two advertisement formats using a two-proportion Z-test.
2. Conduct correlation analysis between:
   - User age and viewing time
   - Viewing time and click behavior
3. Evaluate whether statistical significance leads to meaningful business impact.

Significance level: $\alpha = 0.05$

Methodology:-

1. Collect advertisement interaction data from social media campaigns.
2. Store the dataset in a CSV file.
3. Import the dataset using Python.
4. Separate advertisement data into Format A and Format B groups.
5. Compute click-through rates for both advertisement formats.
6. Apply the Two-Proportion Z-Test to test for significant differences in click-through rates.
7. Calculate the Z-statistic and p-value.
8. Perform correlation analysis between:
   - User age and viewing time
   - Viewing time and click behavior

Implementation

Dataset:-

Ad_Format	Impressions	Clicks	Age	View_Time	Click
A	5000	350	22	5	0
B	4800	420	25	8	1
A	5200	360	23	6	0
B	4900	430	30	7	1
A	5100	340	24	7	1
B	5000	410	27	9	1

Python Code:-

```
import pandas as pd
import numpy as np
from scipy.stats import pearsonr
```

```
from statsmodels.stats.proportion import proportions_ztest

# Load dataset
data = pd.read_csv("advertisement_click_data.csv")

# Separate advertisement formats
adA = data[data['Ad_Format'] == 'A']
adB = data[data['Ad_Format'] == 'B']

# Total clicks and impressions
clicks = np.array([adA['Clicks'].sum(), adB['Clicks'].sum()])
impressions = np.array([adA['Impressions'].sum(), adB['Impressions'].sum()])

# Two proportion Z test
z_stat, p_value = proportions_ztest(clicks, impressions)

print("Z Statistic:", z_stat)
print("P Value:", p_value)

# Correlation analysis
age_view_corr, _ = pearsonr(data['Age'], data['View_Time'])
view_click_corr, _ = pearsonr(data['View_Time'], data['Click'])

print("\nCorrelation between Age and Viewing Time:", age_view_corr)
print("Correlation between Viewing Time and Click Behavior:",
view_click_corr)

# Decision
if p_value < 0.05:
   print("\nReject Null Hypothesis")
   print("There is a significant difference in click-through rates between the
two advertisement formats")
else:
   print("\nFail to Reject Null Hypothesis")
```

Interpretation

The statistical analysis determines whether one advertisement format performs significantly better than the other in terms of click-through rate.
If the p-value obtained from the Two-Proportion Z-Test is less than 0.05, it indicates that the difference in click-through rates between the two advertisement formats is statistically significant.

The correlation analysis helps understand user behavior patterns.
For example:
• A positive correlation between viewing time and clicks indicates that users who spend more time viewing advertisements are more likely to click them.
• A weak or moderate relationship between age and viewing time may indicate that engagement varies slightly across age groups.
These insights help marketing teams design more effective advertisement strategies.

Output:-

```
Z Statistic: -17.298515251459357
P Value: 4.8266850473884295e-67

Correlation between Age and Viewing Time: 0.03393783767140105
Correlation between Viewing Time and Click Behavior: 0.3768876945213388

Reject Null Hypothesis
There is a significant difference in click-through rates between the two advertisement formats
```

Problem Understanding	**Design Document**	**Implementation**	**Output**	**Time Management**	**Viva**	**Total**
10	10	15	5	5	5	50

Result:-

The analysis evaluates the effectiveness of two advertisement formats using statistical testing.

The Two-Proportion Z-Test determines whether the difference in click-through rates between the advertisement formats is statistically significant.